

Advanced Algebra

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Chapter I

Determinants

§1 Matrix Multiplication and Determinants

Question 1.1. Compute the determinant of the following circulant matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{pmatrix} a_1 & a_2 & a_3 & \cdots & a_n \\ a_n & a_1 & a_2 & \cdots & a_{n-1} \\ a_{n-1} & a_n & a_1 & \cdots & a_{n-2} \\ \vdots & \vdots & \vdots & & \vdots \\ a_2 & a_3 & a_4 & \cdots & a_1 \end{pmatrix}$$

Proof. Define the polynomial $f(x) = a_1 + a_2x + a_3x^2 + \cdots + a_nx^{n-1}$, and let $\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_n$ be all the n th roots of unity. Also, let

$$\mathbf{V} = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ \varepsilon_1 & \varepsilon_2 & \varepsilon_3 & \cdots & \varepsilon_n \\ \varepsilon_1^2 & \varepsilon_2^2 & \varepsilon_3^2 & \cdots & \varepsilon_n^2 \\ \vdots & \vdots & \vdots & & \vdots \\ \varepsilon_1^{n-1} & \varepsilon_2^{n-1} & \varepsilon_3^{n-1} & \cdots & \varepsilon_n^{n-1} \end{pmatrix}$$

Then

$$\mathbf{AV} = \begin{pmatrix} f(\varepsilon_1) & f(\varepsilon_2) & f(\varepsilon_3) & \cdots & f(\varepsilon_n) \\ \varepsilon_1 f(\varepsilon_1) & \varepsilon_2 f(\varepsilon_2) & \varepsilon_3 f(\varepsilon_3) & \cdots & \varepsilon_n f(\varepsilon_n) \\ \varepsilon_1^2 f(\varepsilon_1) & \varepsilon_2^2 f(\varepsilon_2) & \varepsilon_3^2 f(\varepsilon_3) & \cdots & \varepsilon_n^2 f(\varepsilon_n) \\ \vdots & \vdots & \vdots & & \vdots \\ \varepsilon_1^{n-1} f(\varepsilon_1) & \varepsilon_2^{n-1} f(\varepsilon_2) & \varepsilon_3^{n-1} f(\varepsilon_3) & \cdots & \varepsilon_n^{n-1} f(\varepsilon_n) \end{pmatrix}$$

Therefore,

$$|\mathbf{A}||\mathbf{V}| = |\mathbf{AV}| = f(\varepsilon_1)f(\varepsilon_2)\cdots f(\varepsilon_n)|\mathbf{V}|$$

Since the ε_i are pairwise distinct, $|\mathbf{V}| \neq 0$, and hence

$$|\mathbf{A}| = f(\varepsilon_1) f(\varepsilon_2) \cdots f(\varepsilon_n)$$

□

Question 1.2. Compute the determinant of the following matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{pmatrix} x & -y & -z & -w \\ y & x & -w & z \\ z & w & x & -y \\ w & -z & y & x \end{pmatrix}$$

1st Proof. Observe that

$$\mathbf{A}\mathbf{A}' = \begin{pmatrix} x & -y & -z & -w \\ y & x & -w & z \\ z & w & x & -y \\ w & -z & y & x \end{pmatrix} \begin{pmatrix} x & y & z & w \\ -y & x & w & -z \\ -z & -w & x & y \\ -w & z & -y & x \end{pmatrix} = \begin{pmatrix} u & 0 & 0 & 0 \\ 0 & u & 0 & 0 \\ 0 & 0 & u & 0 \\ 0 & 0 & 0 & u \end{pmatrix},$$

where $u = x^2 + y^2 + z^2 + w^2$. Therefore,

$$|\mathbf{A}|^2 = (x^2 + y^2 + z^2 + w^2)^4.$$

Setting $x = 1$ and $y = z = w = 0$ in $\mathbf{A}$ gives $|\mathbf{A}| = 1$, so

$$|\mathbf{A}| = (x^2 + y^2 + z^2 + w^2)^2.$$

□

2nd Proof. Let

$$\mathbf{B} = \begin{pmatrix} x & -y \\ y & x \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} z & w \\ w & -z \end{pmatrix}$$

$$\text{Then } |\mathbf{A}| = \begin{vmatrix} \mathbf{B} & -\mathbf{C} \\ \mathbf{C} & \mathbf{B} \end{vmatrix},$$

$$\begin{aligned} |\mathbf{A}| &= |\mathbf{B} + \mathbf{iC}||\mathbf{B} - \mathbf{iC}| \\ &= \begin{vmatrix} x + \mathbf{i}z & -y + \mathbf{i}w \\ y + \mathbf{i}w & x - \mathbf{i}z \end{vmatrix} \begin{vmatrix} x - \mathbf{i}z & -y - \mathbf{i}w \\ y - \mathbf{i}w & x + \mathbf{i}z \end{vmatrix} \\ &= (x^2 + y^2 + z^2 + w^2)^2 \end{aligned}$$

□

§2 The Cauchy–Binet Formula

Theorem 2.1. Let $\mathbf{A} = (a_{ij})$ be an $m \times n$ matrix, let $\mathbf{B} = (b_{ij})$ be an $n \times m$ matrix, and let r be a positive integer with $r \leq m$. Then every minor of order r of $\mathbf{AB}$ satisfies

$$\mathbf{AB} \begin{pmatrix} i_1 & i_2 & \cdots & i_r \\ j_1 & j_2 & \cdots & j_r \end{pmatrix} = \sum_{1 \leq k_1 < \cdots < k_r \leq n} \mathbf{A} \begin{pmatrix} i_1 & i_2 & \cdots & i_r \\ k_1 & k_2 & \cdots & k_r \end{pmatrix} \mathbf{B} \begin{pmatrix} k_1 & k_2 & \cdots & k_r \\ j_1 & j_2 & \cdots & j_r \end{pmatrix}$$

Theorem 2.2. Let $\mathbf{A}$ and $\mathbf{B}$ be $m \times n$ and $n \times m$ matrices, respectively. Prove that the sums of the principal minors of order r of $\mathbf{AB}$ and $\mathbf{BA}$ (the two quantities σ_r) are equal, where $1 \leq r \leq \min\{m, n\}$.

Proof. By the Cauchy–Binet formula,

$$\begin{aligned} & \sum_{1 \leq i_1 < i_2 < \cdots < i_r \leq m} \mathbf{AB} \begin{pmatrix} i_1 & i_2 & \cdots & i_r \\ i_1 & i_2 & \cdots & i_r \end{pmatrix} \\ &= \sum_{1 \leq i_1 < i_2 < \cdots < i_r \leq m} \sum_{1 \leq j_1 < j_2 < \cdots < j_r \leq n} \mathbf{A} \begin{pmatrix} i_1 & i_2 & \cdots & i_r \\ j_1 & j_2 & \cdots & j_r \end{pmatrix} \mathbf{B} \begin{pmatrix} j_1 & j_2 & \cdots & j_r \\ i_1 & i_2 & \cdots & i_r \end{pmatrix} \\ &= \sum_{1 \leq j_1 < j_2 < \cdots < j_r \leq n} \sum_{1 \leq i_1 < i_2 < \cdots < i_r \leq m} \mathbf{B} \begin{pmatrix} j_1 & j_2 & \cdots & j_r \\ i_1 & i_2 & \cdots & i_r \end{pmatrix} \mathbf{A} \begin{pmatrix} i_1 & i_2 & \cdots & i_r \\ j_1 & j_2 & \cdots & j_r \end{pmatrix} \\ &= \sum_{1 \leq j_1 < j_2 < \cdots < j_r \leq n} \mathbf{BA} \begin{pmatrix} j_1 & j_2 & \cdots & j_r \\ j_1 & j_2 & \cdots & j_r \end{pmatrix} \end{aligned}$$

□

Corollary 2.3. $\mathbf{AB}$ and $\mathbf{BA}$ have the same nonzero eigenvalues, counted with multiplicity.

Question 2.4. Let $\mathbf{A}$ and $\mathbf{B}$ be real $m \times n$ matrices. Prove that

$$|\mathbf{AA}'||\mathbf{BB}'| \geq |\mathbf{AB}'|^2$$

Proof. If $m > n$, then $|\mathbf{AA}'| = |\mathbf{BB}'| = |\mathbf{AB}'| = 0$, so the result is immediate. If $m \leq n$, the Cauchy–Binet formula gives

$$\begin{aligned} |\mathbf{AA}'| &= \sum_{1 \leq j_1 < j_2 < \cdots < j_m \leq n} \mathbf{A} \begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix}^2; \\ |\mathbf{BB}'| &= \sum_{1 \leq j_1 < j_2 < \cdots < j_m \leq n} \mathbf{B} \begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix}^2; \\ |\mathbf{AB}'| &= \sum_{1 \leq j_1 < j_2 < \cdots < j_m \leq n} \mathbf{A} \begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix} \mathbf{B} \begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix}, \end{aligned}$$

□

§3 Elementary Block Transformations and Reduction Formulas

Theorem 3.1 (Determinant Reduction Formula). *Let $\mathbf{A}$ be an $m \times m$ matrix, $\mathbf{D}$ an $n \times n$ matrix, $\mathbf{B}$ an $m \times n$ matrix, and $\mathbf{C}$ an $n \times m$ matrix. Prove the following:*

(1) *If $\mathbf{A}$ is invertible, then*

$$\begin{vmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{vmatrix} = |\mathbf{A}| |\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}|$$

(2) *If $\mathbf{D}$ is invertible, then*

$$\begin{vmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{vmatrix} = |\mathbf{D}| |\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}|;$$

(3) *If both $\mathbf{A}$ and $\mathbf{D}$ are invertible, then*

$$|\mathbf{D}| |\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C}| = |\mathbf{A}| |\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}|$$

Corollary 3.2. *Let $\mathbf{A}$ be an $n \times m$ matrix and $\mathbf{B}$ an $m \times n$ matrix. Then*

$$|\lambda \mathbf{I}_n - \mathbf{A}\mathbf{B}| = \lambda^{n-m} |\lambda \mathbf{I}_m - \mathbf{B}\mathbf{A}|$$

Theorem 3.3. *Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. Prove that*

$$\begin{vmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{vmatrix} = |\mathbf{A} + \mathbf{B}| |\mathbf{A} - \mathbf{B}|$$

$$\begin{vmatrix} \mathbf{A} & -\mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{vmatrix} = |\mathbf{A} + \mathbf{i}\mathbf{B}| |\mathbf{A} - \mathbf{i}\mathbf{B}|$$

Corollary 3.4. *If $\mathbf{A}$ and $\mathbf{B}$ are real matrices, then*

$$\begin{vmatrix} \mathbf{A} & -\mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{vmatrix} = |\mathbf{A} + \mathbf{i}\mathbf{B}| |\mathbf{A} - \mathbf{i}\mathbf{B}| = |\det(\mathbf{A} + \mathbf{i}\mathbf{B})|^2 \geq 0$$

Chapter II

Matrices

§1 Special Matrices

Theorem 1.1 (Matrix Units and Standard Unit Vectors). *The standard unit vectors are the following n column vectors in dimension n :*

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

The vectors $e'_1, e'_2, \dots, e'_n$ are called the standard unit row vectors in dimension n . Let $f_1, f_2, \dots, f_m$ be the standard unit column vectors in dimension m .

(1)

$$e'_i e_j = \delta_{ij}$$

$$e_i e'_j = \mathbf{E}_{ij}$$

(2) *If $\mathbf{A} = (a_{ij})$ is an $m \times n$ matrix, then $\mathbf{A}e_i$ is the i th column of $\mathbf{A}$:*

$$(\alpha_1, \alpha_2, \dots, \alpha_n) e_i = \alpha_i$$

Likewise, $f'_i \mathbf{A}$ is the i th row of $\mathbf{A}$:

$$f'_i \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{pmatrix} = \beta_i$$

(3) *If $\mathbf{A} = (a_{ij})$ is an $m \times n$ matrix, then $f'_i \mathbf{A} e_j = a_{ij}$.*

(4)

$$\mathbf{E}_{ij}\mathbf{E}_{kl} = \begin{cases} \mathbf{E}_{il} & , j = k \\ 0 & , j \neq k \end{cases}$$

(5) If $\mathbf{A}$ is an $n \times n$ matrix and $\mathbf{A} = (a_{ij})$, then the i th row of $\mathbf{E}_{ij}\mathbf{A}$ is the j th row of $\mathbf{A}$, while all other rows of $\mathbf{E}_{ij}\mathbf{A}$ are zero.

If $\mathbf{A}$ is an $n \times n$ matrix and $\mathbf{A} = (a_{ij})$, then the j th column of $\mathbf{A}\mathbf{E}_{ij}$ is the i th column of $\mathbf{A}$, while all other columns of $\mathbf{A}\mathbf{E}_{ij}$ are zero.

(6) If $\mathbf{A}$ is an $n \times n$ matrix and $\mathbf{A} = (a_{ij})$, then $\mathbf{E}_{ij}\mathbf{A}\mathbf{E}_{kl} = a_{jk}\mathbf{E}_{il}$.

Theorem 1.2 (Basic Cyclic Matrix). *Let the basic cyclic matrix of order n be*

$$\mathbf{C} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

(1) Circulant matrices are closed under polynomial evaluation, transposition, and inversion (when the inverse exists).

(2) A matrix is circulant if and only if it can be written as $f(\mathbf{C})$ for some $f \in \mathbb{F}[x]$.

(3) The equality $\mathbf{A}\mathbf{C} = \mathbf{C}\mathbf{A}$ holds if and only if $\mathbf{A}$ is circulant.

Theorem 1.3. *Sums, differences, scalar multiples, products (and powers), polynomial expressions, adjugates, and inverses of upper (respectively, lower) triangular matrices remain upper (respectively, lower) triangular. Their diagonal entries are obtained by applying the corresponding operation to the original diagonal entries.*

§2 Invertible Matrices

Theorem 2.1. *Let $\mathbf{A}$ be an $m \times n$ matrix and $\mathbf{B}$ an $n \times m$ matrix. If $\mathbf{I}_m + \mathbf{A}\mathbf{B}$ is invertible, then $\mathbf{I}_n + \mathbf{B}\mathbf{A}$ is also invertible.*

$$(\mathbf{I}_n + \mathbf{B}\mathbf{A})^{-1} = \mathbf{I}_n - \mathbf{B}(\mathbf{I}_m + \mathbf{A}\mathbf{B})^{-1}\mathbf{A}$$

Proof. Observe that $\mathbf{A}(\mathbf{I}_n + \mathbf{B}\mathbf{A}) = (\mathbf{I}_m + \mathbf{A}\mathbf{B})\mathbf{A}$. Hence $(\mathbf{I}_m + \mathbf{A}\mathbf{B})^{-1}\mathbf{A}(\mathbf{I}_n + \mathbf{B}\mathbf{A}) = \mathbf{A}$, and therefore $\mathbf{B}(\mathbf{I}_m + \mathbf{A}\mathbf{B})^{-1}\mathbf{A}(\mathbf{I}_n + \mathbf{B}\mathbf{A}) = \mathbf{B}\mathbf{A}$. It follows that

$$\begin{aligned} \mathbf{I}_n &= \mathbf{I}_n + \mathbf{B}\mathbf{A} - \mathbf{B}\mathbf{A} = (\mathbf{I}_n + \mathbf{B}\mathbf{A}) - \mathbf{B}(\mathbf{I}_m + \mathbf{A}\mathbf{B})^{-1}\mathbf{A}(\mathbf{I}_n + \mathbf{B}\mathbf{A}) \\ &= (\mathbf{I}_n - \mathbf{B}(\mathbf{I}_m + \mathbf{A}\mathbf{B})^{-1}\mathbf{A})(\mathbf{I}_n + \mathbf{B}\mathbf{A}). \end{aligned}$$

Thus $(\mathbf{I}_n + \mathbf{B}\mathbf{A})^{-1} = \mathbf{I}_n - \mathbf{B}(\mathbf{I}_m + \mathbf{A}\mathbf{B})^{-1}\mathbf{A}$.

Corollary 2.2. Let $\mathbf{A}$ and $\mathbf{B}$ be invertible $n \times n$ matrices, and suppose that $\mathbf{A}^{-1} + \mathbf{B}^{-1}$ is invertible. Prove that $\mathbf{A} + \mathbf{B}$ is also invertible and that

$$(\mathbf{A} + \mathbf{B})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1} (\mathbf{A}^{-1} + \mathbf{B}^{-1})^{-1} \mathbf{A}^{-1}$$

Corollary 2.3 (Sherman–Morrison Formula). Let $\mathbf{A}$ be an invertible $n \times n$ matrix, and let α and β be n -dimensional column vectors such that $1 + \beta' \mathbf{A}^{-1} \alpha \neq 0$. Prove that

$$(\mathbf{A} + \alpha \beta')^{-1} = \mathbf{A}^{-1} - \frac{1}{1 + \beta' \mathbf{A}^{-1} \alpha} \mathbf{A}^{-1} \alpha \beta' \mathbf{A}^{-1}$$

§3 Adjugate Matrices

Theorem 3.1. Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix, where $n \geq 2$, and let $\mathbf{A}^*$ be its adjugate. Prove the following statements.

- (1) If $r(\mathbf{A}) = n$, then $r(\mathbf{A}^*) = n$.
- (2) If $r(\mathbf{A}) = n - 1$, then $r(\mathbf{A}^*) = 1$; equivalently, $\mathbf{A}^* = \alpha \beta$.
- (3) If $r(\mathbf{A}) \leq n - 2$, then $r(\mathbf{A}^*) = 0$; equivalently, $\mathbf{A}^* = \mathbf{O}$.

Theorem 3.2. Let $\mathbf{A}_i$ be square matrices. The adjugate of the block diagonal matrix

$$\mathbf{C} = \begin{pmatrix} \mathbf{A}_1 & & & \\ & \mathbf{A}_2 & & \\ & & \ddots & \\ & & & \mathbf{A}_n \end{pmatrix}$$

is

$$\mathbf{C}^* = \begin{pmatrix} \mathbf{A}_1^* \prod_{i \neq 1} |\mathbf{A}_i| & & & \\ & \prod_{i \neq 2} |\mathbf{A}_i| \mathbf{A}_2^* & & \\ & & \ddots & \\ & & & \prod_{i \neq n} |\mathbf{A}_i| \mathbf{A}_n^* \end{pmatrix}$$

Theorem 3.3. The Jordan block

$$\mathbf{J}_n(\lambda) = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & 1 & \\ & & \ddots & \ddots \\ & & & \lambda & 1 \\ & & & & \lambda \end{pmatrix}$$

has adjugate

$$\begin{aligned}
\mathbf{J}_n(\lambda)^* &= |\mathbf{J}_n(\lambda)| \mathbf{J}_n(\lambda)^{-1} \\
&= \lambda^n (\lambda \mathbf{I}_n + \mathbf{J}_n)^{-1} \\
&= \lambda^{n-1} (\mathbf{I}_n - (-\lambda^{-1} \mathbf{J}_n))^{-1} \\
&= \lambda^{n-1} \sum_{k=0}^{n-1} (-\lambda^{-1} \mathbf{J}_n)^k \\
&= \sum_{k=0}^{n-1} (-1)^k \lambda^{n-1-k} \mathbf{J}_n^k
\end{aligned}$$

namely,

$$\begin{pmatrix}
\lambda^{n-1} & -\lambda^{n-2} & & (-1)^{n-2} \lambda & (-1)^{n-1} \\
& \lambda^{n-1} & -\lambda^{n-2} & & (-1)^{n-2} \lambda \\
& & \ddots & \ddots & \\
& & & \lambda^{n-1} & -\lambda^{n-2} \\
& & & & \lambda^{n-1}
\end{pmatrix}$$

§4 Trace

Theorem 4.1. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices and let $k \in \mathbb{F}$. Then

- (1) $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$
- (2) $\text{tr}(k\mathbf{A}) = k \text{tr}(\mathbf{A})$
- (3) $\text{tr}(\mathbf{A}') = \text{tr}(\mathbf{A})$
- (4) If $\mathbf{A}$ is an $m \times n$ matrix and $\mathbf{B}$ is an $n \times m$ matrix, then $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$.
- (5) $\text{tr}(\mathbf{P}^{-1}\mathbf{A}\mathbf{P}) = \text{tr}(\mathbf{A})$

Question 4.2. Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix and define $f(\mathbf{A}) = \sum_{i,j=1}^n a_{ij}^2$. Suppose $\mathbf{P}$ is an invertible $n \times n$ matrix such that $f(\mathbf{PAP}^{-1}) = f(\mathbf{A})$ for every $n \times n$ matrix $\mathbf{A}$. Prove that there exists a nonzero constant c such that $\mathbf{P}'\mathbf{P} = c\mathbf{I}_n$.

Proof. We compute that

$$\begin{aligned}
f(\mathbf{PAP}^{-1}) &= \text{tr}(\mathbf{PAP}^{-1}) \\
&= \text{tr}(\mathbf{PAP}^{-1} (\mathbf{PAP}^{-1})') \\
&= \text{tr}(\mathbf{UAU}^{-1}\mathbf{A}')
\end{aligned}$$

where $\mathbf{U} = \mathbf{P}'\mathbf{P}$. We set

$$\mathbf{A} = \sum_{i,j} a_{ij} \mathbf{E}_{ij}$$

Then

$$\begin{aligned} \text{tr}(\mathbf{U}\mathbf{A}\mathbf{U}^{-1}\mathbf{A}') &= \text{tr}\left(\mathbf{U}\left(\sum_{i,j} a_{ij} \mathbf{E}_{ij}\right)\mathbf{U}^{-1}\left(\sum_{k,l} a_{kl} \mathbf{E}_{kl}\right)\right) \\ &= \text{tr}\left(\sum_{i,j,k,l} a_{ij} a_{kl} \mathbf{U}e_i e_j' \mathbf{U}^{-1} e_k e_l'\right) \\ &= \text{tr}\left(\sum_{i,j,k,l} a_{ij} a_{kl} e_j' \mathbf{U}^{-1} e_k e_l' \mathbf{U} e_i\right) \\ &= \sum_{i,j,k,l} a_{ij} a_{kl} (e_j' \mathbf{U}^{-1} e_k) (e_l' \mathbf{U} e_i) \end{aligned}$$

and

$$\text{tr}(\mathbf{A}'\mathbf{A}) = \sum_{m,n} a_{mn}^2$$

Therefore, we have

$$\sum_{i,j,k,l} a_{ij} a_{kl} (e_j' \mathbf{U}^{-1} e_k) (e_l' \mathbf{U} e_i) = \sum_{m,n} a_{mn}^2$$

for all $(a_{ij})_{n \times n}$. We conclude that $\mathbf{U} = c\mathbf{I}_n$.

§5 Rank

Theorem 5.1. Basic formulas for matrix rank:

- (1) If $k \neq 0$, then $r(k\mathbf{A}) = r(\mathbf{A})$.
- (1) $r(\mathbf{AB}) \leq \min\{r(\mathbf{A}), r(\mathbf{B})\}$
- (2) $r\begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{B} \end{pmatrix} = r(\mathbf{A}) + r(\mathbf{B})$
- (3) $r\begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{B} \end{pmatrix} \geq r(\mathbf{A}) + r(\mathbf{B}), r\begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{D} & \mathbf{B} \end{pmatrix} \geq r(\mathbf{A}) + r(\mathbf{B})$
- (4) $r(\mathbf{A}; \mathbf{B}) \leq r(\mathbf{A}) + r(\mathbf{B}), r\begin{pmatrix} \mathbf{A} \\ \mathbf{B} \end{pmatrix} \leq r(\mathbf{A}) + r(\mathbf{B})$
- (5) $r(\mathbf{A} + \mathbf{B}) \leq r(\mathbf{A}) + r(\mathbf{B}), r(\mathbf{A} - \mathbf{B}) \leq r(\mathbf{A}) + r(\mathbf{B})$
- (6) $r(\mathbf{A} - \mathbf{B}) \geq |r(\mathbf{A}) - r(\mathbf{B})|$

Theorem 5.2 (Frobenius Inequality). Prove that $r(\mathbf{ABC}) \geq r(\mathbf{AB}) + r(\mathbf{BC}) - r(\mathbf{B})$.

Proof. Consider the following elementary block transformations:

$$\begin{pmatrix} \mathbf{ABC} & \mathbf{O} \\ \mathbf{O} & \mathbf{B} \end{pmatrix} \rightarrow \begin{pmatrix} \mathbf{ABC} & \mathbf{AB} \\ \mathbf{O} & \mathbf{B} \end{pmatrix} \rightarrow \begin{pmatrix} \mathbf{O} & \mathbf{AB} \\ -\mathbf{BC} & \mathbf{B} \end{pmatrix} \rightarrow \begin{pmatrix} \mathbf{AB} & \mathbf{O} \\ \mathbf{B} & \mathbf{BC} \end{pmatrix}$$

which give

$$r(\mathbf{ABC}) + r(\mathbf{B}) = r \begin{pmatrix} \mathbf{ABC} & \mathbf{O} \\ \mathbf{O} & \mathbf{B} \end{pmatrix} = r \begin{pmatrix} \mathbf{AB} & \mathbf{O} \\ \mathbf{B} & \mathbf{BC} \end{pmatrix} \geq r(\mathbf{AB}) + r(\mathbf{BC})$$

The result follows.

Corollary 5.3 (Sylvester Inequality). *Let $\mathbf{A}$ be an $m \times n$ matrix and $\mathbf{B}$ an $n \times t$ matrix. Prove that*

$$r(\mathbf{AB}) \geq r(\mathbf{A}) + r(\mathbf{B}) - n$$

Proof. Consider the following elementary block transformations:

$$\begin{pmatrix} \mathbf{I}_n & \mathbf{O} \\ \mathbf{O} & \mathbf{AB} \end{pmatrix} \rightarrow \begin{pmatrix} \mathbf{I}_n & \mathbf{O} \\ \mathbf{A} & \mathbf{AB} \end{pmatrix} \rightarrow \begin{pmatrix} \mathbf{I}_n & -\mathbf{B} \\ \mathbf{A} & \mathbf{O} \end{pmatrix} \rightarrow \begin{pmatrix} \mathbf{B} & \mathbf{I}_n \\ \mathbf{O} & \mathbf{A} \end{pmatrix}$$

which give

$$r(\mathbf{AB}) + n = r \begin{pmatrix} \mathbf{I}_n & \mathbf{O} \\ \mathbf{O} & \mathbf{AB} \end{pmatrix} = r \begin{pmatrix} \mathbf{B} & \mathbf{I}_n \\ \mathbf{O} & \mathbf{A} \end{pmatrix} \geq r(\mathbf{A}) + r(\mathbf{B})$$

Thus $r(\mathbf{AB}) \geq r(\mathbf{A}) + r(\mathbf{B}) - n$.

Corollary 5.4. *If $\mathbf{A}$ is an $m \times n$ matrix, $\mathbf{B}$ is an $n \times t$ matrix, and $\mathbf{AB} = \mathbf{O}$, then $r(\mathbf{A}) + r(\mathbf{B}) \leq n$.*

Corollary 5.5. *Let $\mathbf{A}$ be an $n \times n$ matrix. Then*

$$0 \geq r(\mathbf{A}^{n+2}) - r(\mathbf{A}^{n+1}) \geq r(\mathbf{A}^{n+1}) - r(\mathbf{A}^n) \geq \cdots \geq r(\mathbf{A}) - n$$

Theorem 5.6 (Rank Reduction Formula). *Let $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$ be a block matrix. Prove the following:*

- (1) *If $\mathbf{A}$ is invertible, then $r(\mathbf{M}) = r(\mathbf{A}) + r(\mathbf{D} - \mathbf{CA}^{-1}\mathbf{B})$.*
- (2) *If $\mathbf{D}$ is invertible, then $r(\mathbf{M}) = r(\mathbf{D}) + r(\mathbf{A} - \mathbf{BD}^{-1}\mathbf{C})$.*
- (3) *If both $\mathbf{A}$ and $\mathbf{D}$ are invertible, then $r(\mathbf{A}) + r(\mathbf{D} - \mathbf{CA}^{-1}\mathbf{B}) = r(\mathbf{D}) + r(\mathbf{A} - \mathbf{BD}^{-1}\mathbf{C})$.*

§5.1

Theorem 5.7. *Let $f_1, f_2, \dots, f_k$ be relatively prime polynomials in $\mathbb{F}[x]$, so that $\gcd(f_1, f_2, \dots, f_k) = 1$, and let $\mathbf{A}$ be an $n \times n$ matrix. Then*

$$\sum_{i=1}^k n - r(f_i(\mathbf{A})) = n - r(f(\mathbf{A}))$$

where $f(x) = \prod_{i=1}^k f_i(x)$.

Proof 1. For $k = 2$, perform the following elementary transformations:

$$\begin{aligned} \begin{pmatrix} f_1(\mathbf{A}) & \mathbf{O} \\ \mathbf{O} & f_2(\mathbf{A}) \end{pmatrix} &\rightarrow \begin{pmatrix} f_1(\mathbf{A}) & u_1 f_1(\mathbf{A}) + u_2 f_2(\mathbf{A}) \\ \mathbf{O} & f_2(\mathbf{A}) \end{pmatrix} \rightarrow \begin{pmatrix} f_1(\mathbf{A}) & \mathbf{I}_n \\ \mathbf{O} & f_2(\mathbf{A}) \end{pmatrix} \\ &\rightarrow \begin{pmatrix} f_1(\mathbf{A}) & \mathbf{I}_n \\ -f_1(\mathbf{A})f_2(\mathbf{A}) & \mathbf{O} \end{pmatrix} \rightarrow \begin{pmatrix} \mathbf{O} & \mathbf{I}_n \\ f_1(\mathbf{A})f_2(\mathbf{A}) & \mathbf{O} \end{pmatrix} \end{aligned}$$

This proves the case $k = 2$. Repeated application of this case proves the general statement.

Proof 2 (the kernel-space approach).

$$\ker f(\mathbf{A}) = \oplus \ker f_i(\mathbf{A})$$

Question 5.8. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices over a field $\mathbb{K}$ such that $\mathbf{AB} = \mathbf{BA}$. Prove that

$$r(\mathbf{A} + \mathbf{B}) \leq r(\mathbf{A}) + r(\mathbf{B}) - r(\mathbf{AB})$$

Proof 1 (elementary matrix transformations). Consider the following multiplication of block matrices:

$$\begin{pmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{O} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{B} \end{pmatrix} \begin{pmatrix} \mathbf{I} & -\mathbf{B} \\ \mathbf{I} & \mathbf{A} \end{pmatrix} = \begin{pmatrix} \mathbf{A} + \mathbf{B} & -\mathbf{AB} + \mathbf{BA} \\ \mathbf{B} & \mathbf{BA} \end{pmatrix} = \begin{pmatrix} \mathbf{A} + \mathbf{B} & \mathbf{O} \\ \mathbf{B} & \mathbf{AB} \end{pmatrix}$$

which gives

$$r(\mathbf{A}) + r(\mathbf{B}) = r \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{B} \end{pmatrix} \geq r \begin{pmatrix} \mathbf{A} + \mathbf{B} & \mathbf{O} \\ \mathbf{B} & \mathbf{BA} \end{pmatrix} \geq r(\mathbf{A} + \mathbf{B}) + r(\mathbf{AB}),$$

The result follows.

Proof 2 (the kernel-space approach). Let $V_{\mathbf{A}}$ be the solution space of $\mathbf{A}x = 0$, and define $V_{\mathbf{B}}, V_{\mathbf{AB}}, V_{\mathbf{A+B}}$ analogously. We have $V_{\mathbf{A}} \cap V_{\mathbf{B}} \subseteq V_{\mathbf{A+B}}, V_{\mathbf{A}} \subseteq V_{\mathbf{BA}}$, and $V_{\mathbf{B}} \subseteq V_{\mathbf{AB}}$. Since $\mathbf{AB} = \mathbf{BA}$, we have $V_{\mathbf{BA}} = V_{\mathbf{AB}}$, and hence $V_{\mathbf{A}} + V_{\mathbf{B}} \subseteq V_{\mathbf{AB}}$. Therefore,

$$\dim(V_{\mathbf{A}} \cap V_{\mathbf{B}}) \leq \dim V_{\mathbf{A+B}} = n - r(\mathbf{A} + \mathbf{B})$$

and

$$\dim(V_{\mathbf{A}} + V_{\mathbf{B}}) \leq \dim V_{\mathbf{AB}} = n - r(\mathbf{AB})$$

Adding these inequalities and using the dimension formula for sums and intersections gives

$$\begin{aligned} n - r(\mathbf{A} + \mathbf{B}) + n - r(\mathbf{AB}) &\geq \dim(V_{\mathbf{A}} \cap V_{\mathbf{B}}) + \dim(V_{\mathbf{A}} + V_{\mathbf{B}}) \\ &= \dim V_{\mathbf{A}} + \dim V_{\mathbf{B}} \\ &= n - r(\mathbf{A}) + n - r(\mathbf{B}) \end{aligned}$$

Therefore, $r(\mathbf{A} + \mathbf{B}) + r(\mathbf{AB}) \leq r(\mathbf{A}) + r(\mathbf{B})$, as required.

$$r(\mathbf{A} + \mathbf{B}) \leq r(\mathbf{A}) + r(\mathbf{B}) - r(\mathbf{AB})$$

Proof 3 (the image-space approach). Write the column decompositions $\mathbf{A} = (\alpha_1, \alpha_2, \dots, \alpha_n)$ and $\mathbf{B} = (\beta_1, \beta_2, \dots, \beta_n)$. Let $U_{\mathbf{A}} = L(\alpha_1, \alpha_2, \dots, \alpha_n)$ be the subspace of $\mathbb{K}^n$ spanned by the columns of $\mathbf{A}$, and define $U_{\mathbf{B}}, U_{\mathbf{AB}}, U_{\mathbf{A+B}}$ analogously.

Clearly, $U_{\mathbf{A+B}} \subseteq U_{\mathbf{A}} + U_{\mathbf{B}}$. Since $\mathbf{AB} = (\mathbf{A}\beta_1, \mathbf{A}\beta_2, \dots, \mathbf{A}\beta_n)$, we have $U_{\mathbf{AB}} \subseteq U_{\mathbf{A}}$. Moreover, $\mathbf{AB} = \mathbf{BA}$ implies $U_{\mathbf{AB}} \subseteq U_{\mathbf{A}} \cap U_{\mathbf{B}}$. The preceding inclusions and the dimension formula for sums and intersections now give

$$\begin{aligned} r(\mathbf{A} + \mathbf{B}) + r(\mathbf{AB}) &= \dim U_{\mathbf{A+B}} + \dim U_{\mathbf{AB}} \leq \dim (U_{\mathbf{A}} + U_{\mathbf{B}}) + \dim (U_{\mathbf{A}} \cap U_{\mathbf{B}}) \\ &= \dim U_{\mathbf{A}} + \dim U_{\mathbf{B}} = r(\mathbf{A}) + r(\mathbf{B}). \end{aligned}$$

§5.2

Theorem 5.9. A matrix $\mathbf{A}$ has rank r if and only if it has a nonzero minor $|\mathbf{D}|$ of order r and every bordered minor of order $r + 1$ containing $|\mathbf{D}|$ is zero.

Proof. It suffices to prove sufficiency. Without loss of generality, assume that $|\mathbf{D}|$ is the minor formed by the first r rows and first r columns of $\mathbf{A}$. Write

$$\mathbf{A} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_m \end{pmatrix} = (\beta_1, \beta_2, \dots, \beta_n)$$

for the row and column decompositions of $\mathbf{A}$. Let $\tau_{\leq r}\alpha_i$ denote the truncation of the row vector α_i to its first r entries, and let $\tau_{\leq r}\beta_j$ denote the truncation of the column vector β_j to its first r entries. Since $|\mathbf{D}| \neq 0$, the vectors $\tau_{\leq r}\alpha_1, \dots, \tau_{\leq r}\alpha_r$ are linearly independent, and therefore $\alpha_1, \dots, \alpha_r$ are linearly independent. It remains to prove that they form a maximal linearly independent set among the rows of $\mathbf{A}$.

Suppose, to the contrary, that they are not maximal. We may then add a row, say α_{r+1} , so that $\alpha_1, \dots, \alpha_r, \alpha_{r+1}$ are linearly independent. Let $\mathbf{A}_1$ be the matrix formed by the first $r + 1$ rows of $\mathbf{A}$. Then $\mathbf{A}_1 = (\tau_{\leq r+1}\beta_1, \tau_{\leq r+1}\beta_2, \dots, \tau_{\leq r+1}\beta_n)$ and $r(\mathbf{A}_1) = r + 1$. Since $|\mathbf{D}| \neq 0$, the vectors $\tau_{\leq r}\beta_1, \dots, \tau_{\leq r}\beta_r$ are linearly independent, as are $\tau_{\leq r+1}\beta_1, \dots, \tau_{\leq r+1}\beta_r$. Because $r(\mathbf{A}_1) = r + 1$, some column of $\mathbf{A}_1$, which we may take to be $\tau_{\leq r+1}\beta_{r+1}$, makes $\tau_{\leq r+1}\beta_1, \dots, \tau_{\leq r+1}\beta_r, \tau_{\leq r+1}\beta_{r+1}$ linearly independent. Let $\mathbf{A}_2 = (\tau_{\leq r+1}\beta_1, \dots, \tau_{\leq r+1}\beta_r, \tau_{\leq r+1}\beta_{r+1})$. Thus $\mathbf{A}_2$ is the square submatrix formed by the first $r + 1$ rows and columns of $\mathbf{A}$, and $r(\mathbf{A}_2) = r + 1$. Consequently, $|\mathbf{A}_2| \neq 0$ is a bordered minor of order $r + 1$ containing $|\mathbf{D}|$, contradicting the hypothesis.

Theorem 5.10. Let the rows of an $m \times n$ matrix $\mathbf{A}$ be $\alpha_1, \alpha_2, \dots, \alpha_m$, and suppose $\alpha_{i_1}, \alpha_{i_2}, \dots, \alpha_{i_r}$ form a maximal linearly independent set. Let its columns be $\beta_1, \beta_2, \dots, \beta_n$, and suppose $\beta_{j_1}, \beta_{j_2}, \dots, \beta_{j_r}$

form a maximal linearly independent set. Prove that the submatrix $\mathbf{D}$ at the intersections of these rows and columns satisfies $|\mathbf{D}| \neq 0$.

Proof. Since $\alpha_{i_1}, \alpha_{i_2}, \dots, \alpha_{i_r}$ are maximal linearly independent, every row α_s of $\mathbf{A}$ is a linear combination of them. Let $\tilde{\alpha}_{i_1}, \tilde{\alpha}_{i_2}, \dots, \tilde{\alpha}_{i_r}, \tilde{\alpha}_s$ be the restrictions of the corresponding rows to columns $j_1, j_2, \dots, j_r$. Then every $\tilde{\alpha}_s$ is a linear combination of $\tilde{\alpha}_{i_1}, \tilde{\alpha}_{i_2}, \dots, \tilde{\alpha}_{i_r}$. Now consider $\mathbf{B} = (\beta_{j_1}, \beta_{j_2}, \dots, \beta_{j_r})$, an $m \times r$ matrix of rank r . Every row $\tilde{\alpha}_s$ of $\mathbf{B}$ lies in the span of $\tilde{\alpha}_{i_1}, \tilde{\alpha}_{i_2}, \dots, \tilde{\alpha}_{i_r}$, and the row rank of $\mathbf{B}$ is r . Hence these r rows form a maximal linearly independent set. Thus $\mathbf{D}$ has full rank and $|\mathbf{D}| \neq 0$.

§6

Theorem 6.1. Let $\mathbf{A}$ be an $m \times n$ matrix. Prove the following:

If $r(\mathbf{A}) = n$, so that $\mathbf{A}$ has full column rank, then there exists an $n \times m$ matrix $\mathbf{B}$ of rank n such that $\mathbf{BA} = \mathbf{I}_n$. Such a matrix $\mathbf{B}$ is called a left inverse of $\mathbf{A}$.

If $r(\mathbf{A}) = m$, so that $\mathbf{A}$ has full row rank, then there exists an $n \times m$ matrix $\mathbf{C}$ of rank m such that $\mathbf{AC} = \mathbf{I}_m$. Such a matrix $\mathbf{C}$ is called a right inverse of $\mathbf{A}$.

Proof. (1) Let $\mathbf{P}$ and $\mathbf{Q}$ be invertible matrices of orders m and n , respectively, such that

$$\mathbf{PAQ} = \begin{pmatrix} \mathbf{I}_n \\ \mathbf{O} \end{pmatrix}$$

Therefore, $(\mathbf{I}_n, \mathbf{O}) \mathbf{PAQ} = \mathbf{I}_n$, so $(\mathbf{I}_n, \mathbf{O}) \mathbf{PA} = \mathbf{Q}^{-1}$ and hence $\mathbf{Q}(\mathbf{I}_n, \mathbf{O}) \mathbf{PA} = \mathbf{I}_n$. It suffices to take $\mathbf{B} = \mathbf{Q}(\mathbf{I}_n, \mathbf{O}) \mathbf{P}$. (2) This follows similarly, or by applying part (1) to $\mathbf{A}'$. **Corollary.** A full-column-rank matrix satisfies left cancellation: if $\mathbf{A}$ has full column rank and $\mathbf{AD} = \mathbf{AE}$, then $\mathbf{D} = \mathbf{E}$. Similarly, a full-row-rank matrix satisfies right cancellation: if $\mathbf{A}$ has full row rank and $\mathbf{DA} = \mathbf{EA}$, then $\mathbf{D} = \mathbf{E}$.

Theorem 6.2. Example 3.92 (Full-Rank Factorization). Let the $m \times n$ matrix $\mathbf{A}$ have rank r . Prove that: (1) $\mathbf{A} = \mathbf{BC}$, where $\mathbf{B}$ is an $m \times r$ matrix with $r(\mathbf{B}) = r$ and $\mathbf{C}$ is an $r \times n$ matrix with $r(\mathbf{C}) = r$. Such a factorization is called a full-rank factorization of $\mathbf{A}$. (2) If $\mathbf{A}$ has two full-rank factorizations $\mathbf{A} = \mathbf{B}_1\mathbf{C}_1 = \mathbf{B}_2\mathbf{C}_2$, then there exists an invertible $r \times r$ matrix $\mathbf{P}$ such that $\mathbf{B}_2 = \mathbf{B}_1\mathbf{P}$ and $\mathbf{C}_2 = \mathbf{P}^{-1}\mathbf{C}_1$.

Chapter III

Linear Maps

§1 Eigenvalues and Eigenvectors

Lemma 1.1.

Theorem 1.2. *Let the eigenvalues of an $n \times n$ matrix $\mathbf{A}$ be $\lambda_1, \lambda_2, \dots, \lambda_n$, and let $f(x)$ be a polynomial. Prove that the eigenvalues of $f(\mathbf{A})$ are $f(\lambda_1), f(\lambda_2), \dots, f(\lambda_n)$.*

Theorem 1.3. *Let the eigenvalues of an invertible $n \times n$ matrix $\mathbf{A}$ be $\lambda_1, \lambda_2, \dots, \lambda_n$. Prove that the eigenvalues of $\mathbf{A}^{-1}$ are $\lambda_1^{-1}, \lambda_2^{-1}, \dots, \lambda_n^{-1}$.*

Theorem 1.4. *Let the eigenvalues of an $n \times n$ matrix $\mathbf{A}$ be $\lambda_1, \lambda_2, \dots, \lambda_n$. Prove that the eigenvalues of $\mathbf{A}^*$ are $\overline{\lambda_1}, \overline{\lambda_2}, \dots, \overline{\lambda_n}$.*

§1.1 Reduction Formula

Theorem 1.5 (Eigenvalue Reduction Formula). *Let $\mathbf{A}$ be an $m \times n$ matrix and $\mathbf{B}$ an $n \times m$ matrix, where $m \geq n$. Prove that*

$$|\lambda \mathbf{I}_m - \mathbf{AB}| = \lambda^{m-n} |\lambda \mathbf{I}_n - \mathbf{BA}|$$

§1.2 Eigenvalues and the Coefficients of the Characteristic Polynomial

Theorem 1.6. *Let the characteristic polynomial of an $n \times n$ matrix $\mathbf{A}$ be*

$$f(\lambda) = \lambda^n + a_1 \lambda^{n-1} + \dots + a_{n-1} \lambda + a_n.$$

Prove that

$$(-1)^r a_r = \sum_{1 \leq i_1 < i_2 < \dots < i_r \leq n} \lambda_{i_1} \lambda_{i_2} \dots \lambda_{i_r} = \sum_{1 \leq i_1 < i_2 < \dots < i_r \leq n} \mathbf{A} \begin{pmatrix} i_1 & i_2 & \dots & i_r \\ i_1 & i_2 & \dots & i_r \end{pmatrix}$$

Corollary 1.7. *Let $\mathbf{A}$ be an $n \times n$ matrix. The following three statements are equivalent:*

(1) $\mathbf{A}$ is nilpotent.

(2) Every eigenvalue of $\mathbf{A}$ is 0.

(2) $\text{tr}(\mathbf{A}^k) = 0 \quad (1 \leq k \leq n).$

§2 $\mathcal{L}(M_{m \times n}(\mathbb{K}))$

§2.1 Kronecker Product

Definition 2.1.

Proposition 2.2. Assume that all the following operations are well-defined.

(1) $(\mathbf{A} + \mathbf{B}) \otimes \mathbf{C} = \mathbf{A} \otimes \mathbf{C} + \mathbf{B} \otimes \mathbf{C}, \mathbf{A} \otimes (\mathbf{B} + \mathbf{C}) = \mathbf{A} \otimes \mathbf{B} + \mathbf{A} \otimes \mathbf{C}$

(2) $(k\mathbf{A}) \otimes \mathbf{B} = k(\mathbf{A} \otimes \mathbf{B}) = \mathbf{A} \otimes (k\mathbf{B})$

(3) $(\mathbf{A} \otimes \mathbf{C})(\mathbf{B} \otimes \mathbf{D}) = (\mathbf{AB}) \otimes (\mathbf{CD})$

(4) $(\mathbf{A} \otimes \mathbf{B}) \otimes \mathbf{C} = \mathbf{A} \otimes (\mathbf{B} \otimes \mathbf{C})$

(5) $\mathbf{I}_m \otimes \mathbf{I}_n = \mathbf{I}_{mn}$

(6) $(\mathbf{A} \otimes \mathbf{B})' = \mathbf{A}' \otimes \mathbf{B}'$

(7) If $\mathbf{A}$ and $\mathbf{B}$ are invertible, then $\mathbf{A} \otimes \mathbf{B}$ is invertible and

$$(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$$

(8) If $\mathbf{A}$ is an $m \times m$ matrix and $\mathbf{B}$ is an $n \times n$ matrix, then $|\mathbf{A} \otimes \mathbf{B}| = |\mathbf{A}|^n |\mathbf{B}|^m$.

(9) If $\mathbf{A}$ is an $m \times m$ matrix and $\mathbf{B}$ is an $n \times n$ matrix, then $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = \text{tr}(\mathbf{A}) \cdot \text{tr}(\mathbf{B})$.

Proposition 2.3. Let $\mathbf{A}$ and $\mathbf{B}$ be matrices of orders m and n , respectively. If the eigenvalues of $\mathbf{A}$ are λ_i and those of $\mathbf{B}$ are μ_j , prove that the eigenvalues of $\mathbf{A} \otimes \mathbf{B}$ are $\lambda_i \mu_j$.

Proposition 2.4. If $\mathbf{A}$ and $\mathbf{B}$ are $m \times n$ and $k \times l$ matrices, respectively, then $r(\mathbf{A} \otimes \mathbf{B}) = r(\mathbf{A}) \cdot r(\mathbf{B})$.

Theorem 2.5. Consider the following linear transformation on $\mathcal{L}(M_{m \times n}(\mathbb{K}))$:

$$\varphi(\mathbf{X}) = \mathbf{AXB}$$

Its representing matrix with respect to the basis $\{\mathbf{E}_{1,1}, \dots, \mathbf{E}_{1,n}, \mathbf{E}_{2,1}, \dots, \mathbf{E}_{2,n}, \dots, \mathbf{E}_{m,1}, \dots, \mathbf{E}_{m,n}\}$ is

$$\mathbf{A} \otimes \mathbf{B}'$$

§2.2 Matrix Equation

Theorem 2.6. Let $\mathbf{A}$ and $\mathbf{B}$ be matrices of orders m and n , respectively, and let V be the vector space of all $m \times n$ matrices. Define a linear transformation φ on V by $\varphi(\mathbf{X}) = \mathbf{AX} - \mathbf{XB}$. A representing matrix of φ is

$$\mathbf{A} \otimes \mathbf{I}_n - \mathbf{I}_m \otimes \mathbf{B}'$$

which is similar to the upper triangular matrix

$$\begin{pmatrix} \lambda_1 & * & * & * \\ & \lambda_2 & * & * \\ & & \ddots & \vdots \\ & & & \lambda_m \end{pmatrix} \otimes \mathbf{I}_n - \mathbf{I}_m \otimes \begin{pmatrix} \mu_1 & * & * & * \\ & \mu_2 & * & * \\ & & \ddots & \vdots \\ & & & \mu_n \end{pmatrix}$$

and has eigenvalues $\lambda_i - \mu_j$. Moreover, the following statements are equivalent:

- (1) φ is injective (equivalently, surjective).
- (2) $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = 0$ has only the zero solution.
- (3) For every (equivalently, for some) $\mathbf{C} \in (M_{m \times n}(\mathbb{K}))$, the equation $\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B} = \mathbf{C}$ has a unique solution.

Theorem 2.7. Let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be $m \times m$, $n \times n$, and $m \times n$ matrices, respectively, such that $\mathbf{A}\mathbf{C} = \mathbf{C}\mathbf{B}$ and $\text{r}(\mathbf{C}) = r$. Then $\mathbf{A}$ and $\mathbf{B}$ have at least r eigenvalues in common, counted with multiplicity.

Proof. Let $\mathbf{P}$ and $\mathbf{Q}$ be invertible matrices of orders m and n , respectively, such that

$$\mathbf{P}\mathbf{C}\mathbf{Q} = \begin{pmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}$$

The hypotheses and conclusion are invariant under the equivalence transformations $\mathbf{C} \mapsto \mathbf{P}\mathbf{C}\mathbf{Q}$, $\mathbf{A} \mapsto \mathbf{P}\mathbf{A}\mathbf{P}^{-1}$, and $\mathbf{B} \mapsto \mathbf{Q}^{-1}\mathbf{B}\mathbf{Q}$. Thus we may assume from the outset that $\mathbf{C} = \begin{pmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}$ is in its rank normal form.

Write the corresponding block decompositions as $\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix}$.

Then

$$\mathbf{A}\mathbf{C} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{O} \\ \mathbf{A}_{21} & \mathbf{O} \end{pmatrix}, \quad \mathbf{C}\mathbf{B} = \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}$$

From $\mathbf{A}\mathbf{C} = \mathbf{C}\mathbf{B}$ we obtain $\mathbf{A}_{11} = \mathbf{B}_{11}$, $\mathbf{A}_{21} = \mathbf{O}$, and $\mathbf{B}_{12} = \mathbf{O}$. Hence

$$|\lambda \mathbf{I}_n - \mathbf{A}| = |\lambda \mathbf{I}_r - \mathbf{A}_{11}| \cdot |\lambda \mathbf{I}_{n-r} - \mathbf{A}_{22}|, \quad |\lambda \mathbf{I}_n - \mathbf{B}| = |\lambda \mathbf{I}_r - \mathbf{B}_{11}| \cdot |\lambda \mathbf{I}_{n-r} - \mathbf{B}_{22}|,$$

Therefore, $\mathbf{A}$ and $\mathbf{B}$ have at least r eigenvalues in common, namely the eigenvalues of $\mathbf{A}_{11} = \mathbf{B}_{11}$.

§3 Rational Canonical Form

§3.1 Cyclic Subspaces

Theorem 3.1. Let U be a φ -invariant subspace of V . Then U is cyclic if and only if the matrix of $\varphi|_U$ with respect to some basis of U is the companion matrix of a monic polynomial.

Proof. First prove sufficiency. Suppose the matrix of $\varphi|_U$ with respect to a basis $\{e_1, e_2, \dots, e_r\}$

of U is the companion matrix $\mathbf{C}(d(\lambda))$, where $d(\lambda) = \lambda^r + a_1\lambda^{r-1} + \cdots + a_{r-1}\lambda + a_r$. Then $\varphi(e_i) = e_{i+1}$ ($1 \leq i \leq r-1$) and $\varphi(e_r) = -\sum_{i=1}^r a_{r-i+1}e_i$. Hence $e_i = \varphi^{i-1}(e_1)$ ($2 \leq i \leq r$), and $U = L(e_1, e_2, \dots, e_r) = C(\varphi, e_1)$ is cyclic.

Now prove necessity. Suppose $U = C(\varphi, \alpha)$ is an r -dimensional cyclic subspace. Then $\{\alpha, \varphi(\alpha), \dots, \varphi^{r-1}(\alpha)\}$ is a basis of U . Write

$$\varphi^r(\alpha) = -a_r\alpha - a_{r-1}\varphi(\alpha) - \cdots - a_1\varphi^{r-1}(\alpha)$$

Let $d(\lambda) = \lambda^r + a_1\lambda^{r-1} + \cdots + a_{r-1}\lambda + a_r$. It is immediate that the matrix of $\varphi|_U$ with respect to the basis $\{\alpha, \varphi(\alpha), \dots, \varphi^{r-1}(\alpha)\}$ is the companion matrix $\mathbf{C}(d(\lambda))$. $\square$

Theorem 3.2 (Direct-Sum Decomposition into Cyclic Subspaces). *In general, suppose the λ -matrix of a linear transformation φ over $\mathbb{K}$ is equivalent to $\text{diag}\{1, \dots, 1, d_1(\lambda), \dots, d_k(\lambda)\}$, where each $d_i(\lambda)$ is a nonconstant monic polynomial. This includes, in particular, the cases of invariant factors and elementary (primary) factors. By the theory of rational canonical form, there is a basis of V with respect to which the matrix of φ is*

$$\mathbf{C} = \text{diag}\{\mathbf{C}(d_1(\lambda)), \mathbf{C}(d_2(\lambda)), \dots, \mathbf{C}(d_k(\lambda))\}$$

In this case, V has a direct-sum decomposition into cyclic subspaces:

$$V = C(\varphi, \alpha_1) \oplus C(\varphi, \alpha_2) \oplus \cdots \oplus C(\varphi, \alpha_k),$$

Moreover, the matrix of $\varphi|_{C(\varphi, \alpha_i)}$ with respect to the basis $\{\alpha_i, \varphi(\alpha_i), \dots, \varphi^{r_i-1}(\alpha_i)\}$ is the companion matrix $\mathbf{C}(d_i(\lambda))$, where $r_i = \dim C(\varphi, \alpha_i)$.

Corollary 3.3. *Let φ be a linear transformation on an n -dimensional vector space V over a field $\mathbb{K}$, with characteristic polynomial $f(\lambda)$ and minimal polynomial $m(\lambda)$. The following four statements are equivalent:*

- (1) V is a cyclic space for φ (not necessarily in a unique way).
- (2) The determinantal divisors, or equivalently the invariant factors, of φ are $1, \dots, 1, f(\lambda)$.
- (2') The elementary (primary) factors of φ are $P_1(\lambda)^{r_1}, P_2(\lambda)^{r_2}, \dots, P_k(\lambda)^{r_k}$, where the $P_i(\lambda)$ are distinct monic irreducible polynomials over $\mathbb{K}$.
- (2'') The minimal polynomial $m(\lambda)$ of φ equals its characteristic polynomial $f(\lambda)$.

Corollary 3.4. *Let φ be a linear transformation on an n -dimensional vector space V over a field $\mathbb{K}$, with characteristic polynomial $f(\lambda)$. The following three statements are equivalent:*

- (1) V has only trivial φ -invariant (cyclic) subspaces.
- (1') Every nonzero vector in V is a cyclic vector and makes V a cyclic space.
- (2) $f(\lambda)$ is irreducible over $\mathbb{K}$.

Question 3.5. *Prove that there exists a real $n \times n$ matrix $\mathbf{A}$ satisfying $\mathbf{A}^2 + 2\mathbf{A} + 5\mathbf{I}_n = \mathbf{O}$ if and only if n is even. When $n \geq 4$, show that such a matrix $\mathbf{A}$ has infinitely many invariant subspaces.*

Proof. Necessity: the minimal polynomial satisfies $m(\lambda) \mid g(\lambda)$, while the characteristic polynomial has the form $f(\lambda) = g(\lambda)^k$. Hence $n = \deg f(\lambda) = 2k$ is even.

Sufficiency: let $n = 2k$ be even. The preceding argument shows that the invariant factors of $\mathbf{A}$ are $1, \dots, 1, g(\lambda), \dots, g(\lambda)$, with k copies of $g(\lambda)$. The rational canonical form gives the following matrix satisfying the required condition:

$$\mathbf{A} = \text{diag} \left\{ \begin{pmatrix} 0 & -5 \\ 1 & -2 \end{pmatrix}, \dots, \begin{pmatrix} 0 & -5 \\ 1 & -2 \end{pmatrix} \right\} \quad (k \text{ blocks of order two}).$$

When $n \geq 4$, let $\{e_1, e_2, e_3, e_4\}$ be the first four standard unit column vectors. It is easy to verify that the cyclic subspaces $\{C_l := C(\mathbf{A}, e_1 + le_3) = L(e_1 + le_3, e_2 + le_4), l \in \mathbb{R}\}$ are pairwise distinct $\mathbf{A}$ -invariant subspaces. Thus $\mathbf{A}$ has infinitely many invariant subspaces.

Question 3.6. Let $\mathbf{A}$ be an $n \times n$ matrix over a field $\mathbb{K}$. Prove that if $\text{tr}(\mathbf{A}) = 0$, then $\mathbf{A}$ is similar over $\mathbb{K}$ to a matrix whose diagonal entries are all zero.

Proof. We argue by induction on n . For $n = 1$, $\mathbf{A} = \mathbf{O}$ and the result is immediate.

Assume the result holds for matrices of order less than n , and consider the case of order n . We seek a similarity transformation that produces a zero diagonal entry, after which the induction hypothesis applies. We use rational canonical form. Since both the hypothesis and conclusion are invariant under similarity, we may assume from the outset that $\mathbf{A}$ is in rational canonical form:

$$\mathbf{F} = \text{diag} \{ \mathbf{F}(d_1(\lambda)), \dots, \mathbf{F}(d_k(\lambda)) \}$$

where the $d_i(\lambda)$ are the nonconstant invariant factors of $\mathbf{A}$ and $\deg d_i(\lambda) = r_i$. If every $r_i = 1$, then $\mathbf{A} = \mathbf{O}$ and the result follows. Otherwise, suppose $r_i > 1$ for some i . Interchanging the $(1, 1)$ and (i, i) blocks is a similarity transformation, after which the $(1, 1)$ entry is zero. Thus we may assume that the $(1, 1)$ entry of $\mathbf{A}$ is zero. Write $\mathbf{A} = \begin{pmatrix} 0 & \alpha' \\ \beta & \mathbf{B} \end{pmatrix}$, where $\alpha, \beta \in \mathbb{K}^{n-1}$, $\mathbf{B} \in M_{n-1}(\mathbb{K})$, and $\text{tr}(\mathbf{B}) = 0$. The result now follows from the induction hypothesis.

Corollary 3.7. Let $\mathbf{C}$ be an $n \times n$ matrix over a field $\mathbb{K}$. Prove that there exist $n \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$ over $\mathbb{K}$ such that $\mathbf{AB} - \mathbf{BA} = \mathbf{C}$ if and only if $\text{tr}(\mathbf{C}) = 0$.

§3.2 Polynomial Representations Induced by Commutativity

Theorem 3.8. Let φ be a linear transformation on an n -dimensional vector space V over a field $\mathbb{F}$. Then

$$C(\varphi) = \mathbb{F}[\varphi]$$

holds if and only if the minimal polynomial of φ equals its characteristic polynomial.

Proof. First prove sufficiency. Suppose the minimal polynomial of φ equals its characteristic polynomial $f(\lambda) = \lambda^n + a_1\lambda^{n-1} + \dots + a_{n-1}\lambda + a_n$. Then φ has only one nonconstant invariant factor. By the theory of rational canonical form, there is a basis $\{e_1, e_2, \dots, e_n\}$ of V with respect

to which the matrix of φ is the companion matrix

$$\mathbf{C}(f(\lambda)) = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_n \\ 1 & 0 & \cdots & 0 & -a_{n-1} \\ 0 & 1 & \cdots & 0 & -a_{n-2} \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_1 \end{pmatrix},$$

Thus

$$\varphi(e_1) = e_2, \varphi(e_2) = e_3, \dots, \varphi(e_{n-1}) = e_n, \varphi(e_n) = -a_n e_1 - a_{n-1} e_2 - \cdots - a_1 e_n$$

Let ψ be any linear transformation on V satisfying $\varphi\psi = \psi\varphi$, and write

$$\psi(e_1) = b_n e_1 + b_{n-1} e_2 + \cdots + b_1 e_n$$

Let $g(x) = b_1 x^{n-1} + \cdots + b_{n-1} x + b_n$. We claim that $\psi = g(\varphi)$. For every e_k ,

$$\begin{aligned} \psi(e_k) &= \psi(\varphi^{k-1}(e_1)) = \varphi^{k-1}(\psi(e_1)) = \varphi^{k-1}(g(\varphi)(e_1)) \\ &= g(\varphi)(\varphi^{k-1}(e_1)) = g(\varphi)(e_k) \end{aligned}$$

Since ψ and $g(\varphi)$ agree on every vector in the basis $\{e_1, e_2, \dots, e_n\}$, linearity implies that $\psi = g(\varphi)$.

Now prove necessity. Let the invariant factors of φ be $1, \dots, 1, d_1(\lambda), \dots, d_k(\lambda)$, where the $d_i(\lambda)$ are nonconstant monic polynomials satisfying $d_i(\lambda) \mid d_{i+1}(\lambda)$ for $1 \leq i \leq k-1$. The rational canonical form of φ is $\mathbf{F} = \text{diag}\{\mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_k\}$, where $\mathbf{F}_i = \mathbf{F}(d_i(\lambda))$ is an $n_i \times n_i$ matrix. If the minimal polynomial of φ differs from its characteristic polynomial, then $k \geq 2$. Construct the block diagonal matrix

$$\mathbf{B} = \text{diag}\{\mathbf{I}_{n_1}, \mathbf{O}_{n_2}, \dots, \mathbf{O}_{n_k}\}$$

Clearly, $\mathbf{B}\mathbf{F} = \mathbf{F}\mathbf{B}$. Suppose, for a contradiction, that there is a polynomial $g(x)$ such that $\mathbf{B} = g(\mathbf{F})$; that is,

$$\mathbf{B} = \text{diag}\{g(\mathbf{F}_1), g(\mathbf{F}_2), \dots, g(\mathbf{F}_k)\}$$

Then $g(\mathbf{F}_1) = \mathbf{I}_{n_1}$ and $g(\mathbf{F}_i) = \mathbf{O}$ for $i \geq 2$. Since $d_k(\lambda)$ is the minimal polynomial (and also the characteristic polynomial) of $\mathbf{F}_k$, we have $d_k(\lambda) \mid g(\lambda)$, and hence $d_1(\lambda) \mid g(\lambda)$. Thus $g(\mathbf{F}_1) = \mathbf{O}$, a contradiction. Therefore, $\mathbf{B}$ cannot be expressed as a polynomial in $\mathbf{F}$, and the linear transformation ψ defined by $\mathbf{B}$ has the required property.

§4 Simultaneous Properties Induced by Commutativity

Theorem 4.1. Let φ and ψ be commuting linear transformations on a vector space V over $\mathbb{F}$, so that $\varphi\psi = \psi\varphi$. Then

1. Every eigenspace of φ is invariant under ψ , and every eigenspace of ψ is invariant under φ .
2. $R(\psi)$ is invariant under φ , and $R(\varphi)$ is invariant under ψ .

Corollary 4.2. Let φ and ψ be commuting linear transformations on a vector space V over a field $\mathbb{F}$. If all their eigenvalues lie in $\mathbb{F}$, then φ and ψ have a common eigenvector.

Proof. Choose an eigenvalue $\lambda_0 \in \mathbb{F}$ of φ , and let V_0 be its eigenspace. Then V_0 is invariant under ψ .

Extend a basis of V_0 to a basis of V . With respect to this basis, the matrix of ψ has the block upper triangular form $\begin{pmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{O} & \mathbf{B} \end{pmatrix}$, where $\mathbf{A}$ represents $\psi|_{V_0}$. Hence $|\lambda\mathbf{I}_V - \psi| = |\lambda\mathbf{I} - \mathbf{A}||\lambda\mathbf{I} - \mathbf{B}|$. Since all eigenvalues of ψ lie in $\mathbb{F}$, so do those of $\mathbf{A}$ and of $\psi|_{V_0}$. Choose an eigenvalue $\mu_0 \in \mathbb{F}$ of $\psi|_{V_0}$ with eigenvector $\alpha \in V_0$. Then $\varphi(\alpha) = \lambda_0\alpha$ and $\psi(\alpha) = \mu_0\alpha$, so α is a common eigenvector.

Corollary 4.3. Let $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_m$ be pairwise commuting $n \times n$ matrices over a field $\mathbb{F}$, and suppose all their eigenvalues lie in $\mathbb{F}$. Prove that they have a common eigenvector in $\mathbb{F}^n$.

Proof. We use induction on m . Assume the result holds for fewer than m matrices. Regard all $\mathbf{A}_i$ as linear transformations on the column-vector space $\mathbb{F}^n$. Choose an eigenvalue $\lambda_1 \in \mathbb{F}$ of $\mathbf{A}_1$ and let $V_1 \subseteq \mathbb{F}^n$ be its eigenspace. Then V_1 is invariant under $\mathbf{A}_2, \dots, \mathbf{A}_m$. Their restrictions to V_1 still commute pairwise, and their eigenvalues still lie in $\mathbb{F}$. By the induction hypothesis, $\mathbf{A}_2|_{V_1}, \dots, \mathbf{A}_m|_{V_1}$ have a common eigenvector $\alpha \in V_1$. Since α is also an eigenvector of $\mathbf{A}_1$, it is a common eigenvector of $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_m$.

Theorem 4.4. Let $\mathbf{A}$ be an $n \times n$ matrix over a field $\mathbb{F}$ whose eigenvalues all lie in $\mathbb{F}$. Prove that $\mathbf{A}$ is triangularizable over $\mathbb{F}$; that is, there exists an invertible matrix $\mathbf{P}$ over $\mathbb{F}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is upper triangular.

Proof. We use induction on n . The case $n = 1$ is immediate. Assume the result for matrices of order $n - 1$, and let $\mathbf{A}$ have order n . Choose an eigenvalue $\lambda_1 \in \mathbb{F}$ of $\mathbf{A}$ and an eigenvector $e_1 \in \mathbb{F}^n$ such that $\mathbf{A}e_1 = \lambda_1 e_1$. By the basis-extension theorem, extend e_1 to a basis $\{e_1, e_2, \dots, e_n\}$ of $\mathbb{F}^n$. Then

$$(\mathbf{A}e_1, \mathbf{A}e_2, \dots, \mathbf{A}e_n) = (e_1, e_2, \dots, e_n) \begin{pmatrix} \lambda_1 & * \\ \mathbf{O} & \mathbf{A}_1 \end{pmatrix}$$

where $\mathbf{A}_1$ is an $(n - 1) \times (n - 1)$ matrix over $\mathbb{F}$. Let $\mathbf{P} = (e_1, e_2, \dots, e_n)$. Then $\mathbf{P}$ is invertible and the preceding equality gives $\mathbf{A}\mathbf{P} = \mathbf{P} \begin{pmatrix} \lambda_1 & * \\ \mathbf{O} & \mathbf{A}_1 \end{pmatrix}$, so $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} \lambda_1 & * \\ \mathbf{O} & \mathbf{A}_1 \end{pmatrix}$. It follows that $|\lambda\mathbf{I}_n - \mathbf{A}| = (\lambda - \lambda_1)|\lambda\mathbf{I}_{n-1} - \mathbf{A}_1|$, and hence all eigenvalues of $\mathbf{A}_1$ lie in $\mathbb{F}$. By the induction hypothesis, there exists an invertible $(n - 1) \times (n - 1)$ matrix $\mathbf{Q}$ over $\mathbb{F}$ such that $\mathbf{Q}^{-1}\mathbf{A}_1\mathbf{Q}$ is upper

triangular. Let

$$\mathbf{R} = \mathbf{P} \begin{pmatrix} 1 & \mathbf{O} \\ \mathbf{O} & \mathbf{Q} \end{pmatrix}$$

Then $\mathbf{R}$ is an invertible $n \times n$ matrix over $\mathbb{F}$, and

$$\mathbf{R}^{-1}\mathbf{A}\mathbf{R} = \begin{pmatrix} 1 & \mathbf{O} \\ \mathbf{O} & \mathbf{Q} \end{pmatrix}^{-1} \begin{pmatrix} \lambda_1 & * \\ \mathbf{O} & \mathbf{A}_1 \end{pmatrix} \begin{pmatrix} 1 & \mathbf{O} \\ \mathbf{O} & \mathbf{Q} \end{pmatrix} = \begin{pmatrix} \lambda_1 & * \\ \mathbf{O} & \mathbf{Q}^{-1}\mathbf{A}_1\mathbf{Q} \end{pmatrix}$$

is upper triangular.

Theorem 4.5. Let $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_m$ be pairwise commuting $n \times n$ matrices over a field $\mathbb{F}$, and suppose all their eigenvalues lie in $\mathbb{F}$. Prove that they are simultaneously triangularizable over $\mathbb{F}$; that is, there exists an invertible matrix $\mathbf{P}$ over $\mathbb{F}$ such that $\mathbf{P}^{-1}\mathbf{A}_i\mathbf{P}$ is upper triangular for every $1 \leq i \leq m$.

Theorem 4.6. Let φ and ψ be linear transformations on an n -dimensional vector space V over a field $\mathbb{F}$. Suppose $\varphi\psi = \psi\varphi$ and both φ and ψ are diagonalizable. Prove that they are simultaneously diagonalizable.

Proof. We use induction on $\dim V$. The case $n = 1$ is immediate. Assume the result for vector spaces of dimension less than n . Let the distinct eigenvalues of φ be $\lambda_1, \dots, \lambda_s \in \mathbb{F}$, with corresponding eigenspaces $V_1, \dots, V_s$. Since φ is diagonalizable,

$$V = V_1 \oplus \dots \oplus V_s.$$

If $s = 1$, then $\varphi = \lambda_1 \mathbf{I}_V$ is a scalar transformation. Choose a basis of V in which the matrix of ψ is diagonal; in the same basis, the matrix of φ is $\lambda_1 \mathbf{I}_n$. If $s > 1$, then $\dim V_i < n$, and every V_i is invariant under ψ . The restrictions $\varphi|_{V_i}$ and $\psi|_{V_i}$ commute and are both diagonalizable. By the induction hypothesis, they are simultaneously diagonalizable on V_i . Combining these bases of the V_i gives a basis of V in which both φ and ψ are diagonal.

Theorem 4.7. Let $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_m$ be pairwise commuting $n \times n$ matrices over a field $\mathbb{F}$, each diagonalizable over $\mathbb{F}$. Prove that they are simultaneously diagonalizable over $\mathbb{F}$.

§5 Criteria for Diagonalizability

§5.1 Existence of n Linearly Independent Eigenvectors

§5.2 Existence of a Complete System of Eigenvectors

A linear map or matrix is diagonalizable if and only if it has a complete system of eigenvectors; equivalently, the algebraic multiplicity of every eigenvalue equals its geometric multiplicity.

Question 5.1. Suppose $\mathbf{A}$ is an $m \times n$ matrix, $\mathbf{B}$ is an $n \times m$ matrix, and $|\mathbf{AB}| \neq 0$. Then $\mathbf{AB}$ is diagonalizable if and only if $\mathbf{BA}$ is diagonalizable.

Proof.



Chapter IV

Canonical Forms

§1 Construction of Canonical Forms

Theorem 1.1. *Given matrices $\mathbf{A}$ and $\mathbf{B}$ over a field $\mathbb{F}$. Then $\mathbf{A}$ and $\mathbf{B}$ are similar if and only if $\lambda \mathbf{I}_n - \mathbf{A}$ and $\lambda \mathbf{I}_n - \mathbf{B}$ have the same Smith normal form; equivalently, they are equivalent in $M_n(\mathbb{F}[\lambda])$.*

Proposition 1.2. *Suppose φ is a linear transformation on a finite-dimensional vector space V over a field $\mathbb{F}$, with characteristic polynomial $f(\lambda)$ and minimal polynomial $m(\lambda)$.*

$$f(\lambda) = P_1(\lambda)^{r_1} P_2(\lambda)^{r_2} \cdots P_k(\lambda)^{r_k}, m(\lambda) = P_1(\lambda)^{s_1} P_2(\lambda)^{s_2} \cdots P_k(\lambda)^{s_k},$$

where $P_i(\lambda)$ are distinct monic irreducible polynomials over $\mathbb{F}$, and $r_i \geq s_i \geq 1$ for all i .

1. $V = \bigoplus \ker(P_i(\varphi)^{s_i})$
2. $\varphi|_{V_i}$ has minimal polynomial $P_i(\lambda)^{s_i}$ and characteristic polynomial $P_i(\lambda)^{r_i}$, where $V_i = \ker(P_i(\varphi)^{s_i})$.